



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology | Department of Computer Science

FINAL TERM PROJECT REPORT

Heart Disease Prediction Using Machine Learning

Course Title: Programming in Python

Section: B

Semester: Summer 2025-2026

Submitted By:

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Date of Submission: 7th Sep, 2026

GitHub Link: <https://github.com/sami4007/Heart-Disease-Prediction-Using-Machine-Learning>

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1. Project Title and Group Information

- **Project Title:** Heart Disease Prediction Using Machine Learning
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2. Problem Statement, Stakeholders, Objective, and Success Criteria

2.1 Problem Statement

Heart disease is one of the major health-related challenges, and early identification of patients at risk can support timely medical attention. Traditional assessment generally depends on clinical measurements and expert interpretation, which may require considerable time and experience.

This project aims to develop a machine learning-based classification system that predicts whether a patient is likely to have heart disease based on selected clinical and demographic features. The project evaluates multiple machine learning models and compares their performance to identify the most effective predictive approach.

2.2 Stakeholders

- **Healthcare Professionals:** Can use predictive results as an additional decision-support reference during patient assessment.
- **Healthcare Researchers:** Can use the dataset and model results to analyze relationships between clinical features and heart disease.
- **Patients:** May benefit from earlier identification of potential risk through appropriate clinical assessment.
- **Machine Learning Practitioners:** Can use the project as an example of applying classification algorithms to healthcare data.

2.3 Project Objective & Unit of Analysis

The primary objective of this project is to develop and evaluate a machine learning pipeline for binary heart disease prediction.

The model uses patient-related clinical features such as age, sex, chest pain type, resting blood pressure, cholesterol, maximum heart rate, exercise-induced angina, oldpeak, and ST slope to predict the target class.

The target variable contains two classes:

- 0 = No Heart Disease
- 1 = Heart Disease

The project compares a majority-class baseline, Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), TabPFN, TabICLv2, and Mitra to evaluate conventional and modern approaches for binary heart disease classification.

2.4 Success Criteria

The models are evaluated using the following performance metrics:

- **Accuracy** — Overall percentage of correct predictions.
- **Precision** — Proportion of predicted positive cases that are actually positive.
- **Recall** — Proportion of actual positive cases correctly identified.
- **F1-Score** — Harmonic mean of Precision and Recall.
- **ROC-AUC** — Measures the model's ability to distinguish between the two target classes.

For this healthcare prediction task, Recall is particularly important, because failing to identify a patient with potential heart disease can be more concerning than generating an additional false positive.

3. Dataset Source, License, Data Dictionary, and Ethical Considerations

3.1 Dataset Name: Heart Disease Dataset

3.2 Source: Kaggle

3.3 Dataset URL: <https://www.kaggle.com/datasets/eishkaran/heart-disease>

The Heart Disease dataset was obtained from Kaggle and used for developing and evaluating the machine learning models in this project. The dataset contains clinical and demographic attributes that are used to predict the presence or absence of heart disease.

Kaggle

Target Variable: target

- 0 = No Heart Disease
- 1 = Heart Disease

3.4 Data Dictionary

Feature	Type	Description
age	Numerical	Age of the patient
sex	Categorical	Gender of the patient
chest pain type	Categorical	Type of chest pain experienced
resting bp s	Numeric	Resting blood pressure
cholesterol	Numeric	Serum cholesterol level
fasting blood sugar	Numeric	Fasting blood sugar measurement
resting ecg	Categorical	Resting electrocardiogram result
max heart rate	Numerical	Maximum heart rate achieved
exercise angina	Categorical	Exercise-induced angina
oldpeak	Numerical	ST depression induced by exercise
ST slope	Categorical	Slope of the peak exercise ST segment
target	Binary	0 = No Heart Disease, 1 = Heart Disease

3.5 Ethical Considerations

This project uses patient-related clinical data for academic machine learning research and prediction. The model is intended only as a **decision-support tool** and should not be considered a replacement for professional medical diagnosis. Model predictions should be interpreted carefully, particularly because incorrect predictions may have medical consequences.

4. Data Audit, Exploratory Analysis, and Visualization Findings

4.1 Data Audit Summary

The Heart Disease Dataset used in this project contains clinical and physiological attributes for patient-level binary classification. The raw CSV file contains 1,190 records and 12 columns, with 11 predictor variables and one target variable. The target variable, "target", is encoded as 0 for no heart disease and 1 for the presence of heart disease. The initial audit found no explicit null values in the raw dataset. However, 272 exact duplicate records were identified. These duplicates were removed before further analysis, resulting in a cleaned dataset of 918 records. The audit also identified one zero value in resting blood pressure and 172 zero values in cholesterol. Because zero is not a meaningful physiological measurement for either variable in this context, these entries were treated as invalid values and replaced with missing values before median imputation.

Audit Item	Result	Action / Interpretation
Original dataset	1,190 rows × 12 columns	Raw dataset loaded from CSV
Missing values	0 explicit null values	No null entries in the raw file
Duplicate records	272 (22.86%)	Removed before modeling
Cleaned dataset	918 rows × 12 columns	Dataset used for analysis
Resting blood pressure = 0	1 record	Treated as invalid and median-imputed
Cholesterol = 0	172 records	Treated as invalid and median-imputed
Target classes	0 = No disease; 1 = Disease	Binary classification target

4.2 Exploratory Data Analysis Findings

Following duplicate removal, the cleaned dataset contains 410 observations in class 0 and 508 observations in class 1. Class 0 therefore represents 44.69% of the data, while class 1 represents 55.31%. The difference between the two classes is moderate and does not constitute a severe imbalance, although the majority-class baseline must still be considered when interpreting accuracy.

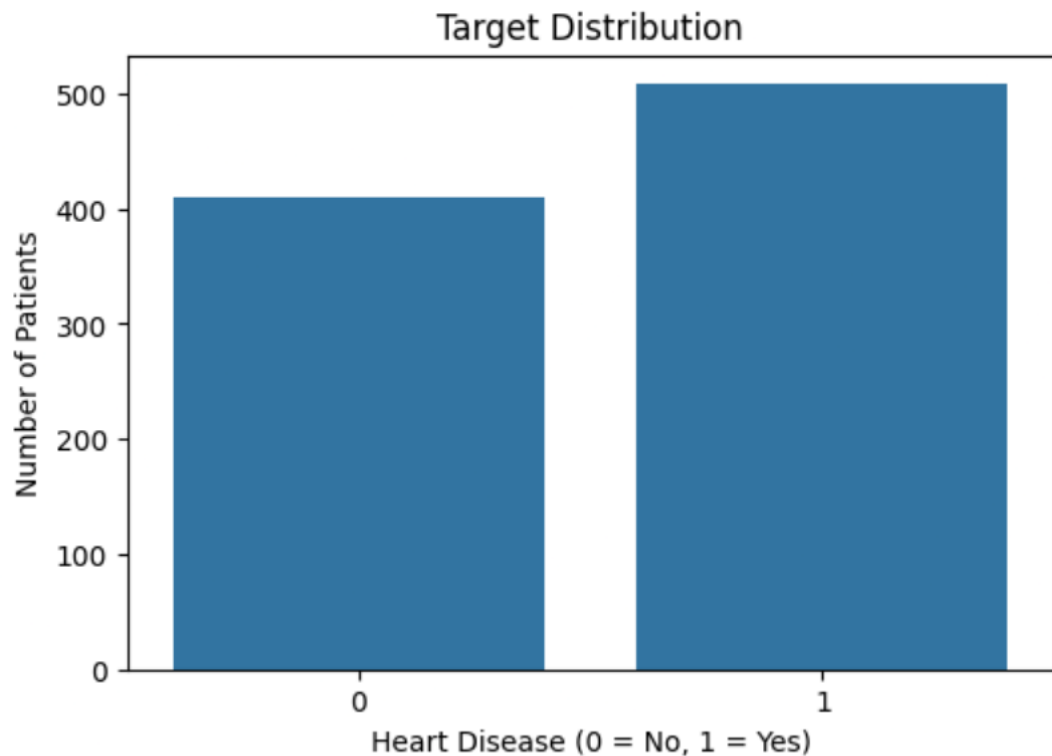


Figure 4.1: Target Distribution

Description: The target distribution shows that the cleaned dataset contains more patients with heart disease than without heart disease. The two classes remain sufficiently represented for stratified model development, while the moderate class difference makes a majority-class baseline necessary for meaningful evaluation.

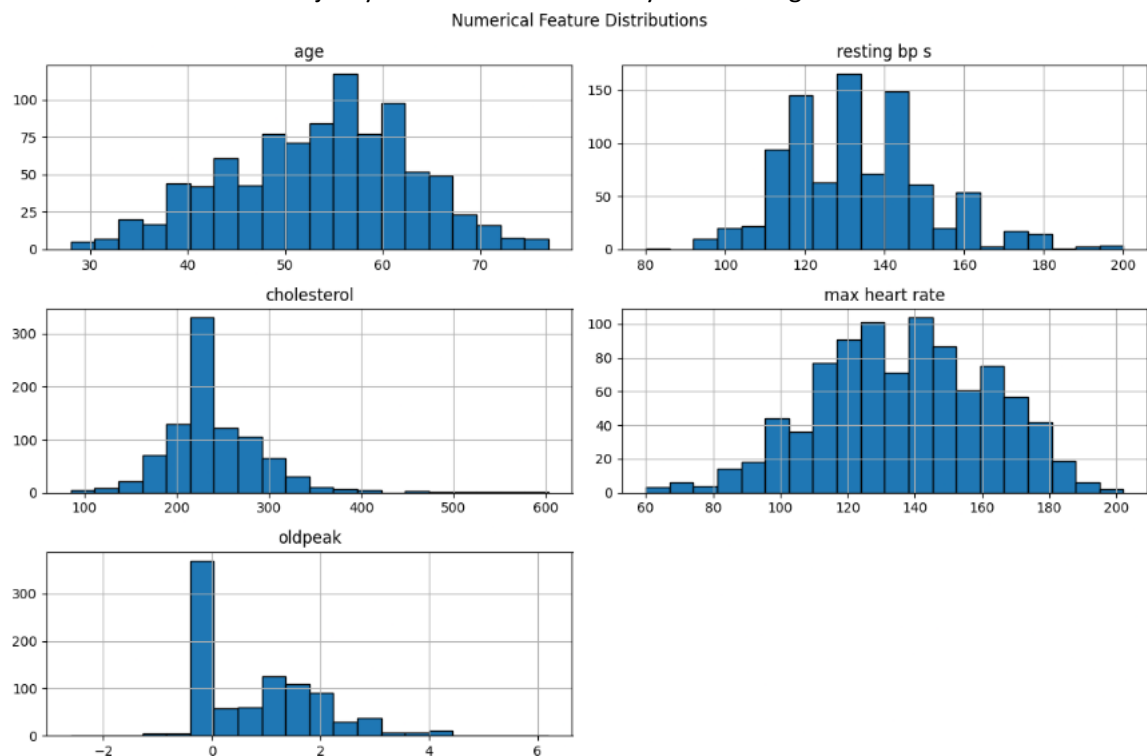


Figure 4.2: Numerical Feature Distributions

Description: The numerical-feature histograms show the distributions of age, resting blood pressure, cholesterol, maximum heart rate, and oldpeak. Age is broadly distributed across middle-aged and older patients. Cholesterol and oldpeak show noticeable concentration and right-skewed behavior, while maximum heart rate spans a comparatively wide

range. These distributions justify scaling the numerical variables before distance- and coefficient-based modeling.

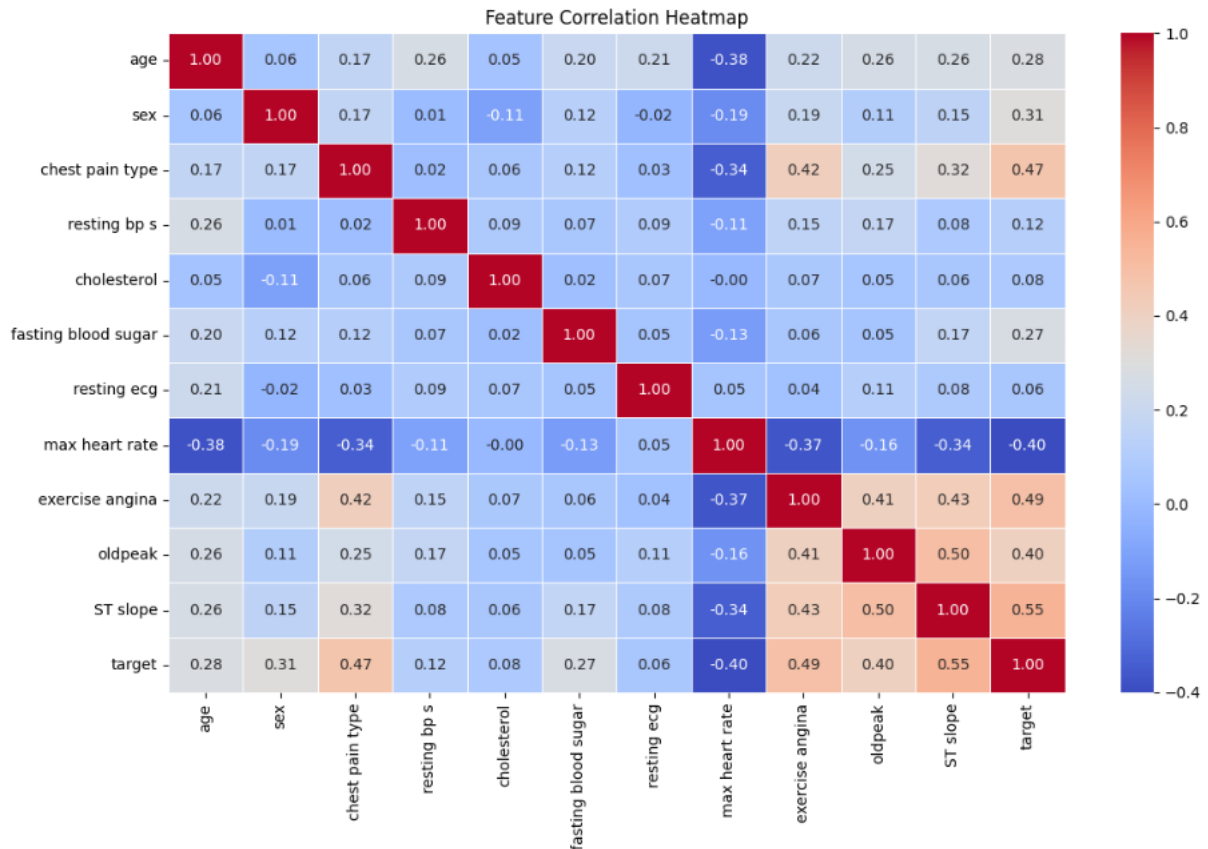


Figure 4.3: Feature Correlation Heatmap

Description: The correlation heatmap provides an overview of linear relationships among the encoded variables. The target has its strongest positive associations with ST slope (0.55), exercise-induced angina (0.49), and chest pain type (0.47), while maximum heart rate shows a negative association with the target (-0.40). Oldpeak also shows a positive association (0.40). These relationships indicate that several clinical variables contain useful information for distinguishing the two target classes.

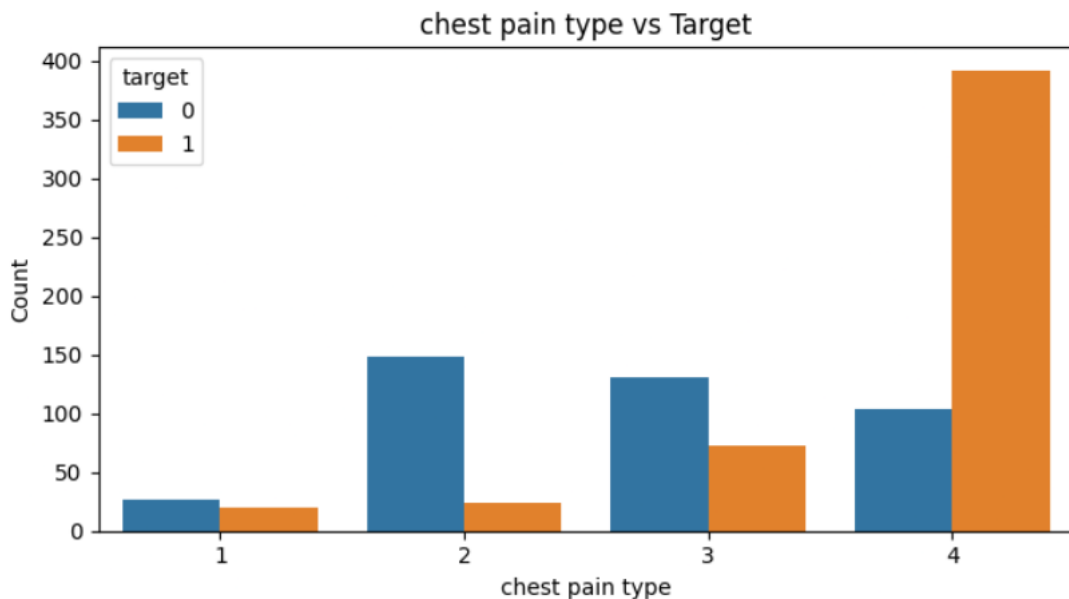


Figure 4.4: Chest Pain Type vs Target

Description: The distribution of chest pain categories differs substantially between the two target classes. In particular, chest pain type 4 is much more prevalent among patients classified with heart disease, whereas types 2 and 3 show comparatively larger counts in the no-disease group.

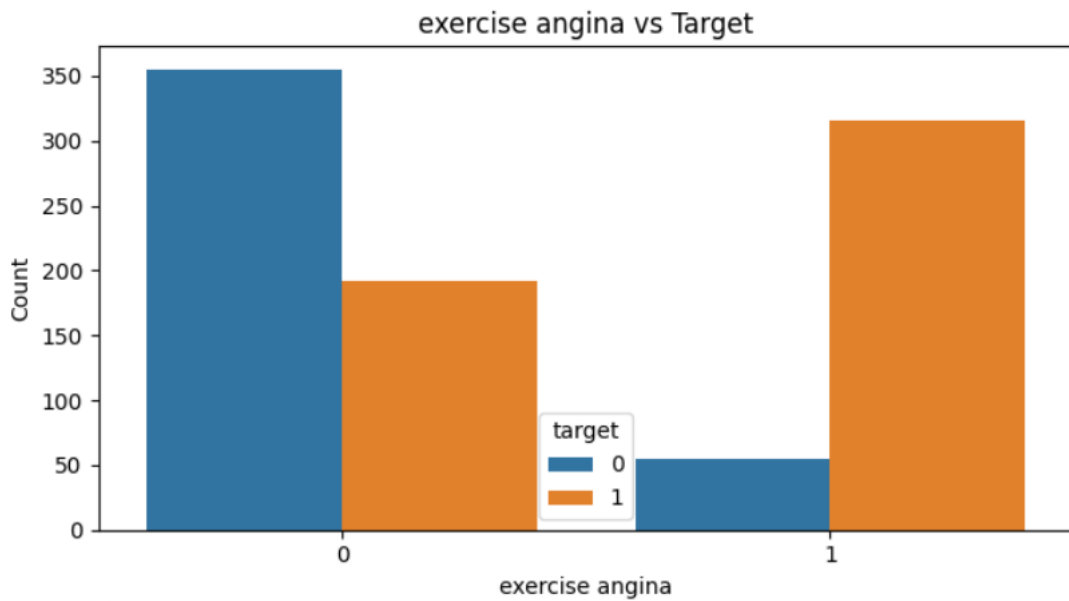


Figure 4.5: Exercise-Induced Angina vs Target

Description: Exercise-induced angina shows a clear difference between the two target classes. Patients with exercise-induced angina (value 1) are predominantly found in the heart disease class, whereas the no-angina category contains a larger proportion of patients without heart disease.

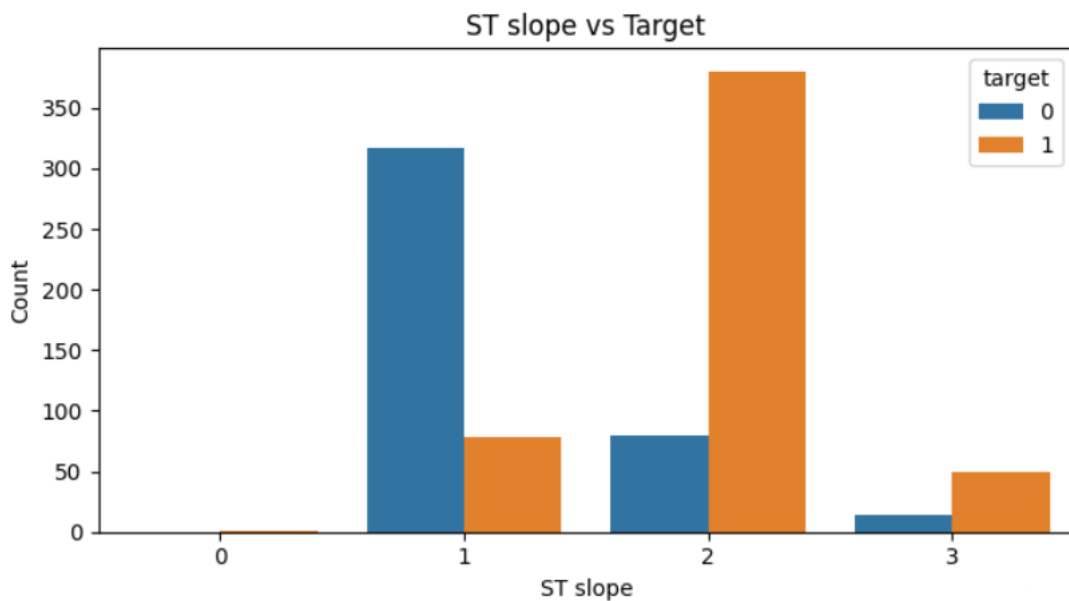


Figure 4.6: ST Slope vs Target

Description: ST slope demonstrates one of the clearest target-wise differences in the EDA. ST slope category 2 is strongly represented among patients with heart disease, while category 1 is more common among patients without heart disease.

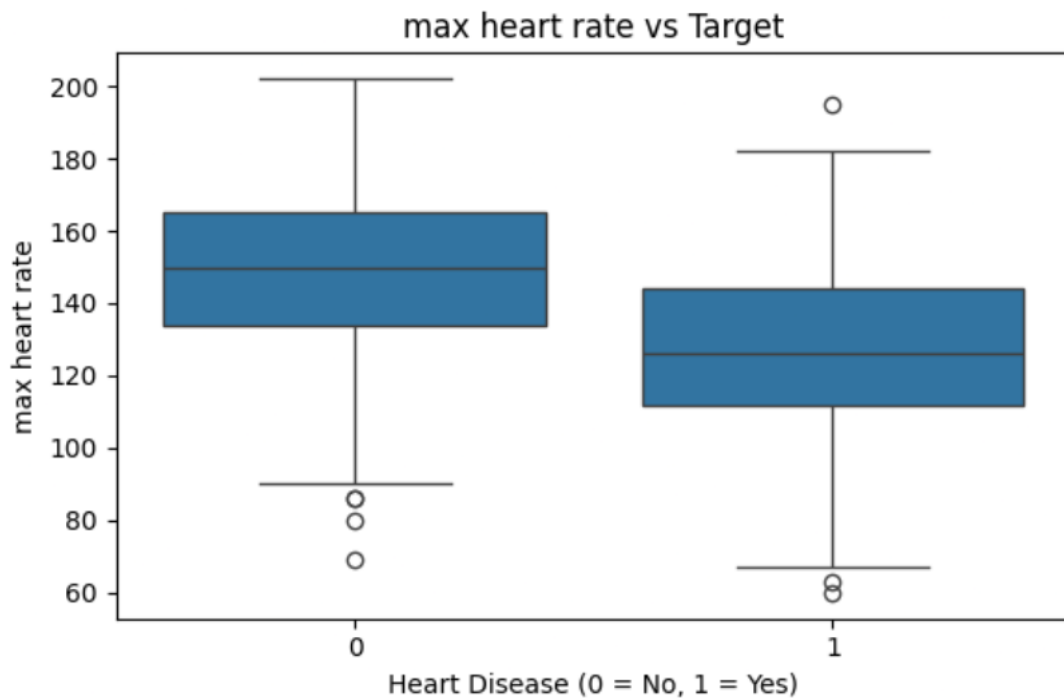


Figure 4.7: Maximum Heart Rate vs Target

Description: The boxplots indicate a lower central tendency for maximum heart rate among patients with heart disease compared with patients without heart disease. This visual difference is consistent with the negative correlation between maximum heart rate and the target observed in the heatmap.

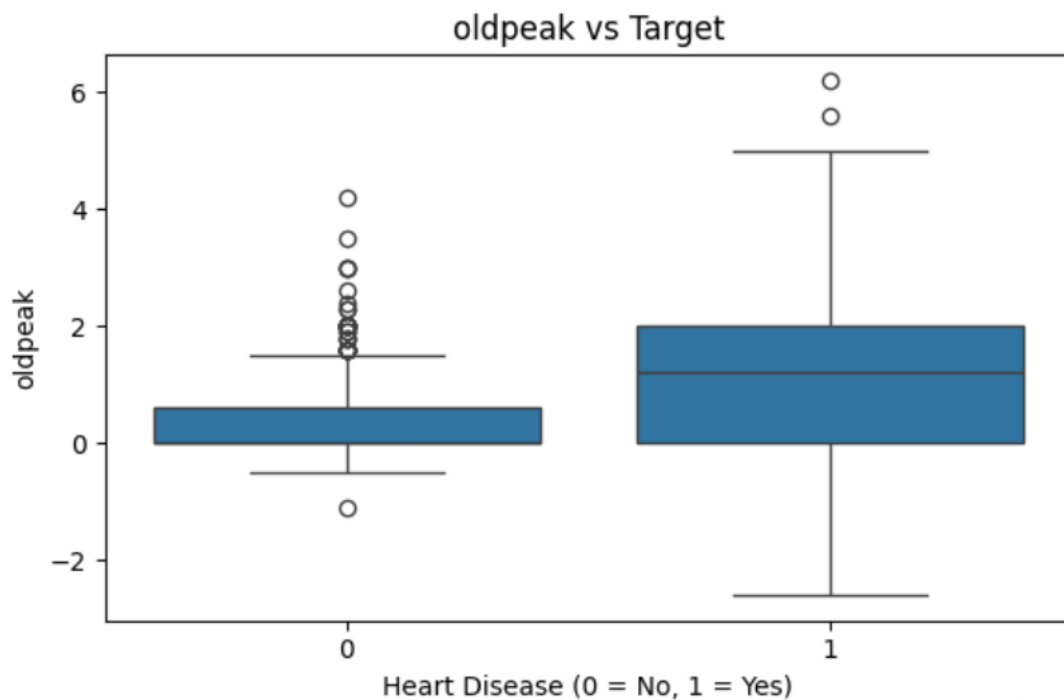


Figure 4.8: Oldpeak vs Target

Description: Old peak shows a higher median and wider spread in the heart disease group. The difference between the two groups supports its inclusion as a potentially informative numerical predictor.

Overall, the EDA indicates that the dataset contains several variables with visible differences between the two target classes. The strongest exploratory patterns are observed in ST slope, exercise-induced angina, chest pain type, maximum heart rate, and old peak. These findings support proceeding with supervised binary classification rather than relying on the target distribution alone.

5. Split Strategy, Preprocessing Decisions, and Leakage Controls

5.1 Split Strategy

After duplicate removal and invalid-value treatment, the dataset was separated into the predictor matrix X and the binary target vector y . The notebook uses an 80/20 stratified train-test split with `random_state=42`. This produces 734 training observations and 184 testing observations.

Partition	Samples	Class 0	Class 1
Training set	734	328	406
Testing set	184	82	102

Stratification preserves the target distribution across the two partitions. The training set contains 44.69% class 0 and 55.31% class 1, while the test set contains 44.57% class 0 and 55.43% class 1. The test set is kept separate from model fitting and is used for final performance assessment.

5.2 Preprocessing Decisions

- Exact duplicate records were removed before model development.
- Zero values in resting blood pressure and cholesterol were treated as invalid measurements.
- Invalid values were converted to missing values and replaced using median imputation.
- Five numerical variables—age, resting blood pressure, cholesterol, maximum heart rate, and oldpeak—were standardized using `StandardScaler`.
- Six categorical variables—sex, chest pain type, fasting blood sugar, resting ECG, exercise-induced angina, and ST slope—were encoded using `OneHotEncoder(handle_unknown='ignore', sparse_output=False)`.
- The Column Transformer produced 22 processed features from the original 11 predictors.

5.3 Leakage Control

The notebook performs the train-test split after the data-cleaning step. The `StandardScaler` and `OneHotEncoder` in Step 8 are then fitted only on X_{train} and subsequently used to transform X_{test} . This prevents the test set from contributing to the learned scaling parameters or categorical encoding mappings.

There is, however, an important methodological limitation in the current implementation. The medians used to replace invalid resting blood pressure and cholesterol values are calculated in Step 5 before the train-test split. Therefore, the imputation statistics are derived from the complete cleaned dataset rather than from the training partition alone. This means that the current notebook is leakage-controlled for scaling and encoding, but the imputation stage is not strictly training-only.

For a fully leakage-safe implementation, the split should occur before the imputation statistics are calculated. The training-set medians should be learned from X_{train} only and then applied unchanged to X_{test} . This distinction is important when reporting the methodology because test-set information should not influence any learned preprocessing parameter.

5.4 Reproducibility Settings

A fixed random state of 42 is used throughout the notebook for reproducibility. The stratified split therefore produces the same training and testing partitions whenever the workflow is rerun under the same data and software environment.

6. Baseline Definition and Result

A majority-class baseline was established using `DummyClassifier(strategy='most_frequent')`. The baseline does not learn relationships among the clinical variables; instead, it predicts the most frequent class for every patient. Because class 1 is the majority class in the cleaned dataset, the baseline predicts heart disease for every test observation.

Metric	Baseline Result
Accuracy	0.5543
Precision	0.5543
Recall	1.0000
F1-Score	0.7133
ROC-AUC	0.5000

The baseline achieves an accuracy of 0.5543 because approximately 55.43% of the test observations belong to the majority class. Its recall for class 1 is 1.0000 because every observation is classified as having heart disease. This apparently high recall is misleading when considered alone: every patient without heart disease is also incorrectly classified as

positive. Consequently, the baseline has no meaningful discriminative ability, reflected by its ROC-AUC of 0.5000. The F1-score of 0.7133 provides a useful benchmark for subsequent models, but it should not be interpreted as evidence of a useful predictive system.

6.1 Baseline Confusion Pattern

On the 184-patient test set, the majority-class baseline predicts all 184 cases as class 1. The resulting confusion pattern is 0 true negatives, 82 false positives, 102 true positives, and 0 false negatives. This demonstrates why accuracy alone is insufficient for this task and establishes a deliberately simple benchmark against which the subsequent machine-learning models can be compared.

7. Candidate Models, Reasoning, Hyperparameters, and Tuning

7.1 Candidate Model Selection

Seven classification approaches were evaluated in this project: a majority-class baseline, Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), TabPFN, TabICLv2, and Mitra. The models were selected to compare a simple reference model, commonly used machine-learning algorithms, a distance-based method, and newer approaches for tabular data.

Model	Reason for Inclusion	Notebook Configuration
Baseline	Majority-class reference is used to determine whether the trained models provide useful predictive performance.	strategy='most_frequent'
Logistic Regression	Linear classification model used as a simple and interpretable reference.	max_iter=1000; random_state=42
Random Forest	Tree-based ensemble model capable of learning non-linear relationships and feature interactions.	n_estimators=200; random_state=42
KNN	Distance-based model that makes predictions based on the similarity between observations.	n_neighbors=5
TabPFN	Modern tabular-learning approach evaluated on the structured clinical dataset.	device='cpu'
TabICLv2	Additional modern tabular-learning approach used for comparison with TabPFN and classical models.	device='cpu'; random_state=42
Mitra	Additional tabular-learning approach included as part of the model comparison.	fine_tune=False; fit_weighted_ensemble=False; eval_metric='roc_auc'

7.2 Validation and Model Selection

Model validation was performed using StratifiedKFold with 5 folds, shuffling enabled, and random_state=42. Stratification was used to maintain a similar distribution of the two target classes across the validation folds. The primary model-selection criterion was mean cross-validation ROC-AUC. ROC-AUC was selected because it measures how well the model separates the two classes across different classification thresholds. Accuracy, precision, recall, and F1-score were also recorded for comparison.

Model	CV Accuracy	CV Precision	CV Recall	CV F1	CV ROC-AUC
TabPFN	0.8679	0.8676	0.9016	0.8837	0.9419
TabICLv2	0.8679	0.8682	0.9016	0.8837	0.9397
Mitra	0.8624	0.8636	0.8967	0.8790	0.9340
Random Forest	0.8583	0.8620	0.8893	0.8745	0.9261
Logistic Regression	0.8379	0.8538	0.8572	0.8547	0.9090
KNN	0.8433	0.8505	0.8743	0.8614	0.8957
Baseline	0.5531	0.5531	1.0000	0.7123	0.5000

TabPFN achieved the highest mean cross-validation ROC-AUC of 0.9419, followed by TabICLv2 with 0.9397 and Mitra with 0.9340. TabPFN also achieved the highest mean cross-validation F1-score, tied with TabICLv2 at 0.8837. Based on the predefined ROC-AUC selection criterion, TabPFN was selected as the final model.

7.3 Hyperparameter Tuning

The notebook does not implement an explicit GridSearchCV or randomized hyperparameter search for these models. Instead, the configurations shown above were used directly, while 5-fold stratified cross-validation was used to compare the resulting models and select the final approach. Therefore, the reported configurations should be described as fixed model settings rather than as the result of an automated hyperparameter optimization procedure.

7.4 Methodological Validation Note

The current notebook contains two preprocessing-related limitations that should be kept in mind when interpreting the validation results. First, invalid zero values are converted to missing values and median-imputed before the train/test split, so the imputation statistics are derived from the complete cleaned dataset. Second, the cross-validation runs for models using X_train processed reuse a preprocessing transformation fitted before the CV folds rather than refitting the transformation independently inside each fold. The final test set remains separate from model fitting, but these choices can make the corresponding validation estimates somewhat optimistic.

8. Final Metrics, Diagnostic Plots, and Computational Cost

8.1 Untouched Test Set Evaluation

The results below are the final test-set metrics produced by the notebook.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
KNN	0.8804	0.9000	0.8824	0.8911	0.9385
TabPFN	0.8967	0.8952	0.9216	0.9082	0.9370
TabICLv2	0.8967	0.8807	0.9412	0.9100	0.9341
Mitra	0.8913	0.8727	0.9412	0.9057	0.9269
Random Forest	0.8804	0.8846	0.9020	0.8932	0.9223
Logistic Regression	0.8587	0.8455	0.9118	0.8774	0.9055
Baseline	0.5543	0.5543	1.0000	0.7133	0.5000

TabPFN achieved an accuracy of 0.8967, precision of 0.8952, recall of 0.9216, F1-score of 0.9082, and ROC-AUC of 0.9370 on the final test set. TabICLv2 achieved the same accuracy of 0.8967, while obtaining a higher recall of 0.9412 and F1-score of 0.9100. KNN produced a slightly higher final-test ROC-AUC of 0.9385 than TabPFN's 0.9370. However, TabPFN remained the selected model because the model-selection decision was based on the 5-fold cross-validation ROC-AUC rather than the final test-set result. TabPFN achieved a CV ROC-AUC of 0.9419, compared with 0.8957 for KNN.

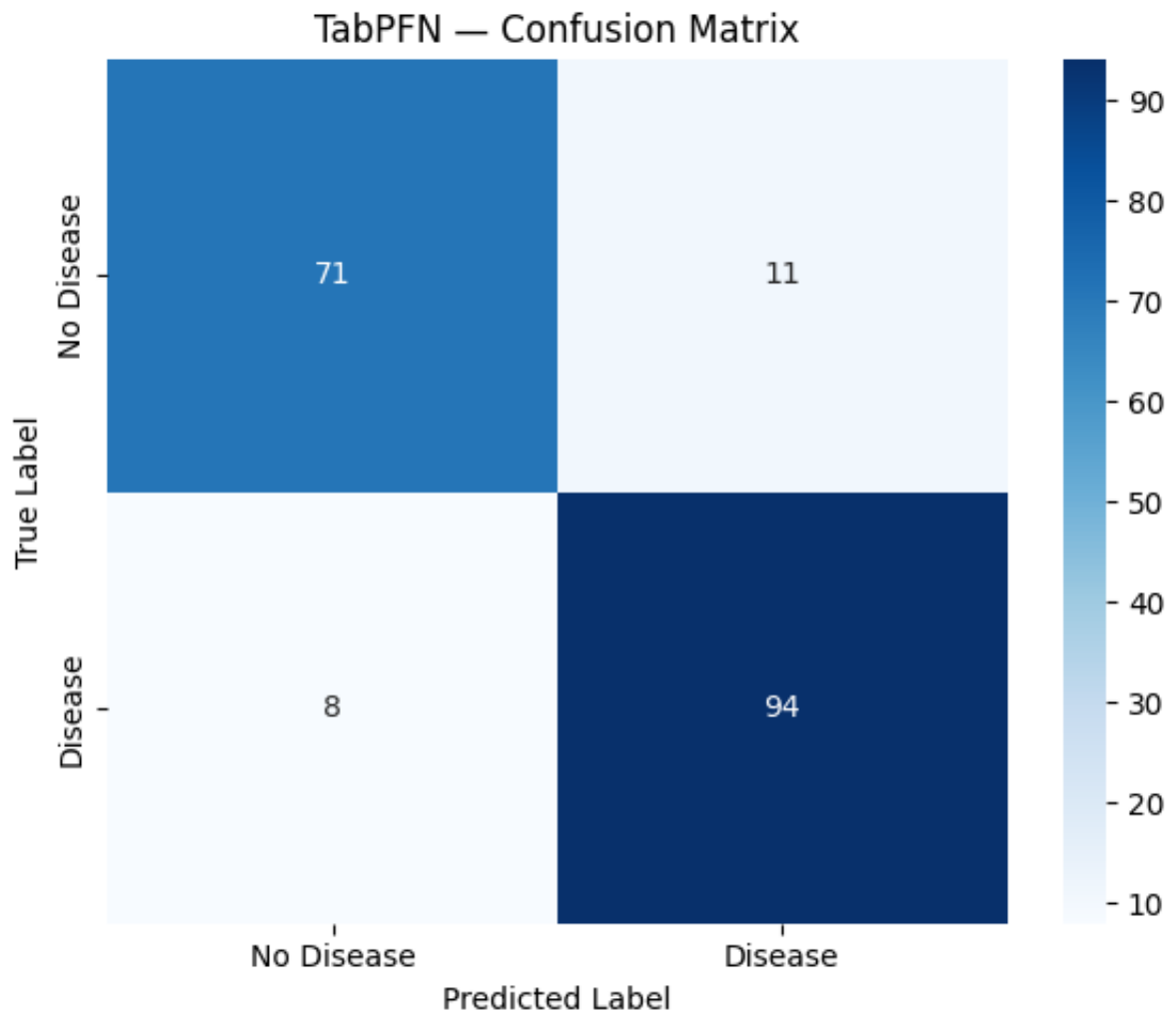


Figure 8.1: TabPFN Confusion Matrix

Description: The confusion matrix shows 71 true negatives, 11 false positives, 8 false negatives, and 94 true positives. Therefore, the model correctly classified 165 out of the 184 test samples. The number of false negatives is important because these represent disease cases that were predicted as no disease. The model produced 8 false negatives, resulting in a recall of 92.16% for the positive class.

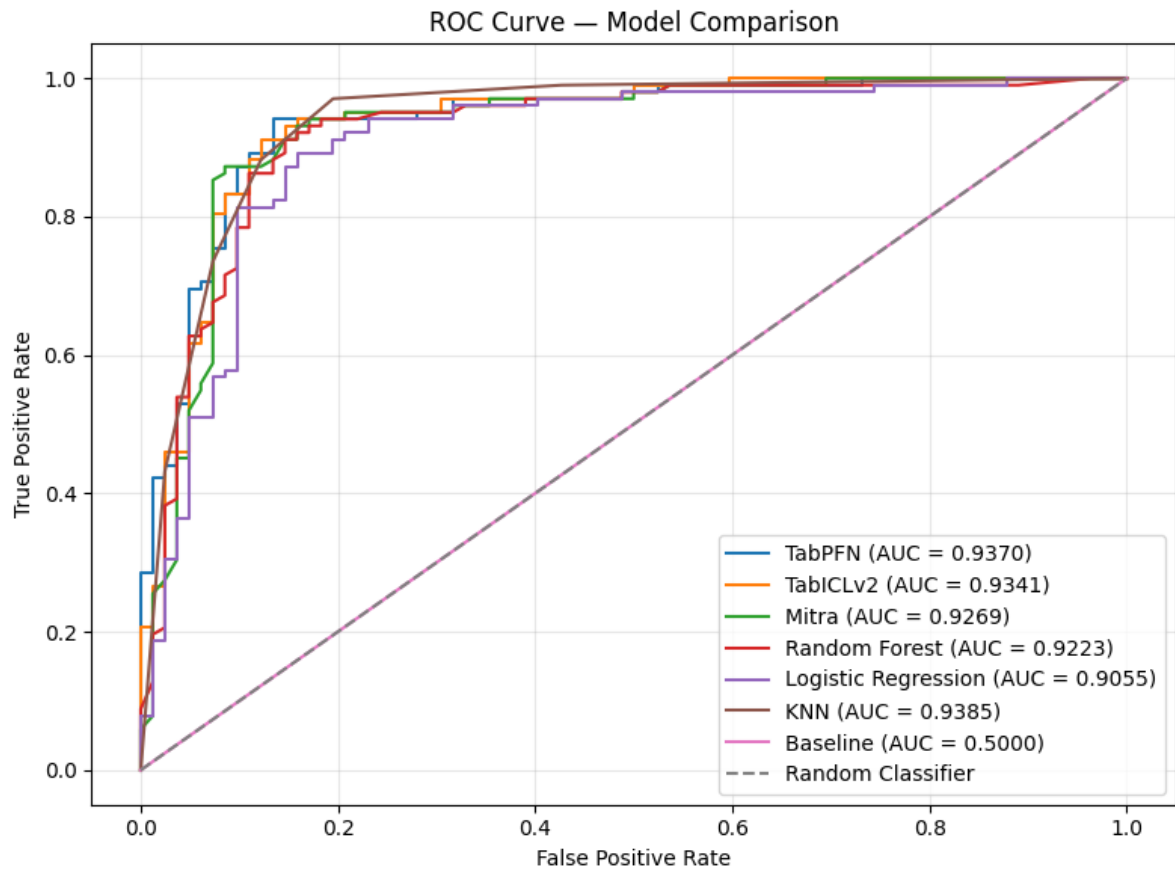


Figure 8.2: ROC Curve — Model Comparison

Description: The ROC curve compares the true-positive rate and false-positive rate of the evaluated models at different classification thresholds. The curves for the trained models remain substantially above the random-classifier line. On the final test set, KNN achieved the highest ROC-AUC of 0.9385, followed by TabPFN with 0.9370 and TabICLv2 with 0.9341. Logistic Regression and Random Forest also performed above the baseline, while the majority-class baseline obtained an ROC-AUC of 0.5000.

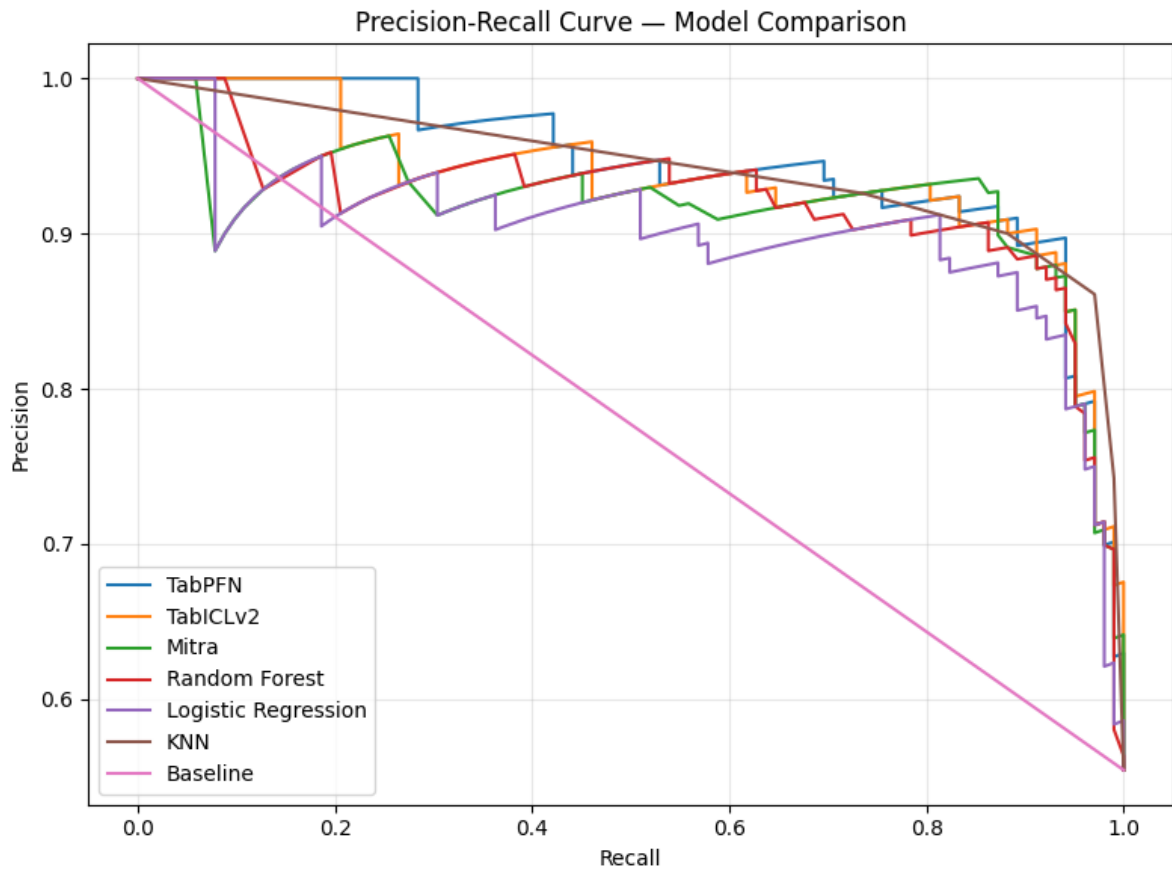


Figure 8.3: Precision-Recall Curve — Model Comparison

Description: The Precision-Recall curves show the relationship between precision and recall at different classification thresholds. The trained models show considerably better precision-recall performance than the majority-class baseline. This curve provides an additional view of model performance because both precision and recall are relevant to the classification of heart disease cases.

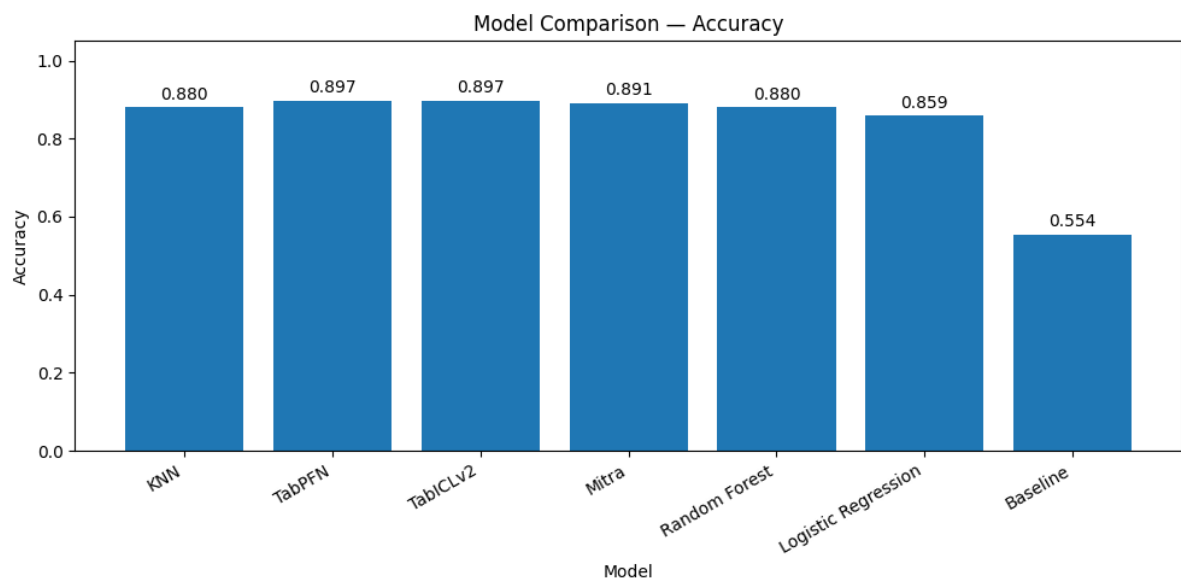


Figure 8.4: Model Comparison — Accuracy

Description: The accuracy comparison shows that TabPFN and TabICLv2 achieved the highest test accuracy, with both models obtaining 0.8967. Mitra achieved an accuracy of 0.8913, while Random Forest and KNN both obtained 0.8804. Logistic Regression achieved 0.8587, and the majority-class baseline achieved 0.5543.

8.2 Diagnostic Interpretation

The confusion matrix indicates that the selected TabPFN model correctly classified most of the observations in the test set. It correctly identified 94 of the 102 positive cases and 71 of the 82 negative cases. The ROC curve shows that the evaluated machine-learning models provide good class separation compared with the majority-class baseline. KNN obtained the highest test ROC-AUC, although TabPFN was selected using the cross-validation results. The Precision-Recall curve also shows that the trained models provide a better balance between precision and recall than the baseline. Overall, the results indicate that the machine-learning models learned useful patterns from the clinical features and performed substantially better than the simple majority-class reference.

8.3 Computational Cost

The notebook provides reproducibility information such as the Python and library versions and records that the models were run on CPU for TabPFN and TabICLv2. However, the workflow does not explicitly record model training time or peak memory usage. Therefore, no numerical training-time or memory-cost comparison is reported rather than introducing unsupported values.

9. Error Analysis, Imbalance, Assumptions, Bias, and Limitations

9.1 Error Analysis of the Selected Model

Error analysis was performed using the predictions of the selected TabPFN model on the final test set. The test set contained 184 observations, of which 165 were correctly classified and 19 were incorrectly classified. The 19 incorrect predictions consisted of 11 false positives and 8 false negatives. The resulting overall error rate was **10.33%**.

Error Category	Count	Interpretation
Correct predictions	165	Predicted class matched the actual target.
Incorrect predictions	19	Predicted class did not match the actual target.
False positives	11	No-disease cases predicted as disease.
False negatives	8	Disease cases predicted as no disease.
Overall error rate	10.33%	19 incorrect predictions among 184 test samples.

The false-negative count is lower than the false-positive count for TabPFN, which is consistent with its recall of 92.16%. The notebook also records the individual misclassified test samples together with their clinical features, actual labels, predicted labels, and prediction probabilities. These records are saved by the workflow for further inspection.

9.2 Class Imbalance

After the data-cleaning process, the dataset contained 410 observations in class 0 and 508 observations in class 1. This corresponds to approximately 44.69% class 0 and 55.31% class 1. The difference between the two classes is relatively small, so the dataset does not show severe class imbalance. However, the classes are not perfectly balanced. For this reason, accuracy was not used as the only evaluation metric. Precision, recall, F1-score, ROC-AUC, and the confusion matrix were also considered.

The majority-class baseline was included to show the performance that could be obtained by always predicting the most frequent class.

9.3 Assumptions and Potential Bias

The analysis assumes that the selected clinical features contain enough information to distinguish between the two target classes and that the target values correctly represent the classification outcome. The dataset may not represent all patient populations or healthcare settings. Differences in patient characteristics and data collection may affect the patterns learned by the models. The presence of duplicate records and invalid zero values may also affect the results. Therefore, the model performance may differ when applied to other populations or independent datasets.

9.4 Limitations and Responsible Use

The dataset is relatively small compared with the diversity of real-world clinical populations. The results therefore should not be interpreted as evidence of clinical effectiveness. In addition, the model predictions are intended for machine-learning project evaluation and should not be treated as a medical diagnosis. External validation on independent clinical data, assessment across relevant patient groups, and appropriate clinical oversight would be required before any real-world medical application.

10. Reproducibility and Computational Documentation

10.1 Reproducibility

The train/test split used an 80/20 stratified split with `random_state=42`, and the cross-validation procedures used stratified 5-fold validation with shuffling and a fixed random state.

The notebook was developed and executed in Google Colab using Python. The main environment versions recorded during the project include **Python 3.13.15**, **NumPy 2.1.3**, **Pandas 2.2.3**, **Scikit-learn 1.6.1**, **Matplotlib 3.10.0**, and **Seaborn 0.13.2**.

The notebook is organized in a sequential order, starting from data loading and data-quality checks and ending with model evaluation, interpretation, and error analysis. The final notebook was run from a clean Colab session to check that the workflow executes in order.

10.2 Running the Project

The project can be reproduced using the following general steps:

- Open the submitted .ipynb notebook in Google Colab.
- Install the required Python packages if they are not already available.
- Upload or provide access to the permitted heart disease dataset CSV file.
- Run the notebook cells sequentially from the first section to the final evaluation section.
- For TabPFN, provide the required authentication key when prompted.

The notebook generates the main tables, evaluation metrics, confusion matrix, ROC/PR curves, feature-importance results, and error-analysis results.

The project files are organized around the notebook, README, requirements/environment information, and the dataset.

10.3 Computational Cost

The project mainly compares model performance rather than hardware efficiency. The models include conventional machine-learning methods and three pretrained tabular models: TabPFN, TabICLv2, and Mitra. Their computational requirements can differ depending on the Google Colab environment and model configuration.

Therefore, the reported performance results should be interpreted together with the model complexity and computational requirements rather than based only on the highest metric.

11. Conclusion, Practical Interpretation, and Responsible Use

11.1 Conclusion

This project developed a complete machine-learning workflow for classifying heart disease risk using a public tabular dataset. The workflow included data auditing, exploratory analysis, preprocessing, model training, cross-validation, final evaluation, feature interpretation, and error analysis.

Seven approaches were evaluated: a majority-class baseline, Logistic Regression, Random Forest, KNN, TabPFN, TabICLv2, and Mitra.

Based on 5-fold cross-validation ROC-AUC, TabPFN achieved the highest score of 0.9419 and was selected as the final model. On the held-out test set, it achieved 89.67% accuracy, 92.16% recall, 90.82% F1-score, and 0.9370 ROC-AUC.

11.2 Practical Interpretation

The results show that pretrained tabular models can perform competitively on this heart-disease dataset. TabPFN correctly classified most test observations while maintaining a good balance between precision and recall.

Its confusion matrix contained 71 true negatives, 11 false positives, 8 false negatives, and 94 true positives. False negatives are particularly important because they represent high-risk cases predicted as lower-risk.

The feature-importance analyses identified ST slope, chest pain type, exercise angina, oldpeak, maximum heart rate, and cholesterol as influential features.

11.3 Responsible Use and Limitations

This project is intended for academic and educational purposes only and should not be used as a medical diagnostic tool.

The dataset may not represent all populations, so the model's performance may not generalize to new patients or healthcare settings. The preprocessing and validation limitations discussed earlier may also make some estimates slightly optimistic.

Because false negatives can be more concerning in health-risk screening, recall and false-negative behaviour should be considered alongside accuracy and ROC-AUC.

12. References and Contribution Statement

12.1 References

1. **eishkaran**. *Heart Disease Dataset*. Kaggle.
<https://www.kaggle.com/datasets/eishkaran/heart-disease>
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12.2 Contribution Statement

The project was completed as a group, with each member contributing to the development and evaluation of different parts of the machine-learning workflow.

Group Member	Main Contribution
Saif Al Sami	Developed and evaluated the TabPFN model, including its training, validation, test evaluation, confusion matrix, ROC/PR analysis, and feature-importance analysis.
Md. Rezoyanul Islam Ananno	Developed and evaluated Logistic Regression, Random Forest & KNN and contributed to the overall model comparison and notebook integration.
Benazir Newaz Khan	Developed and evaluated Mitra model and contributed to the final project integration and analysis.
Md. Nazrul Islam Nabil	Developed and evaluated TabICLv2 model and contributed to the final project integration and analysis.